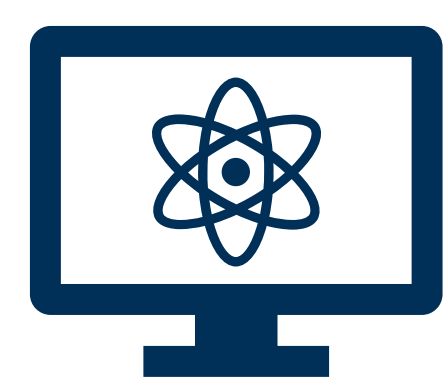


Project Overview

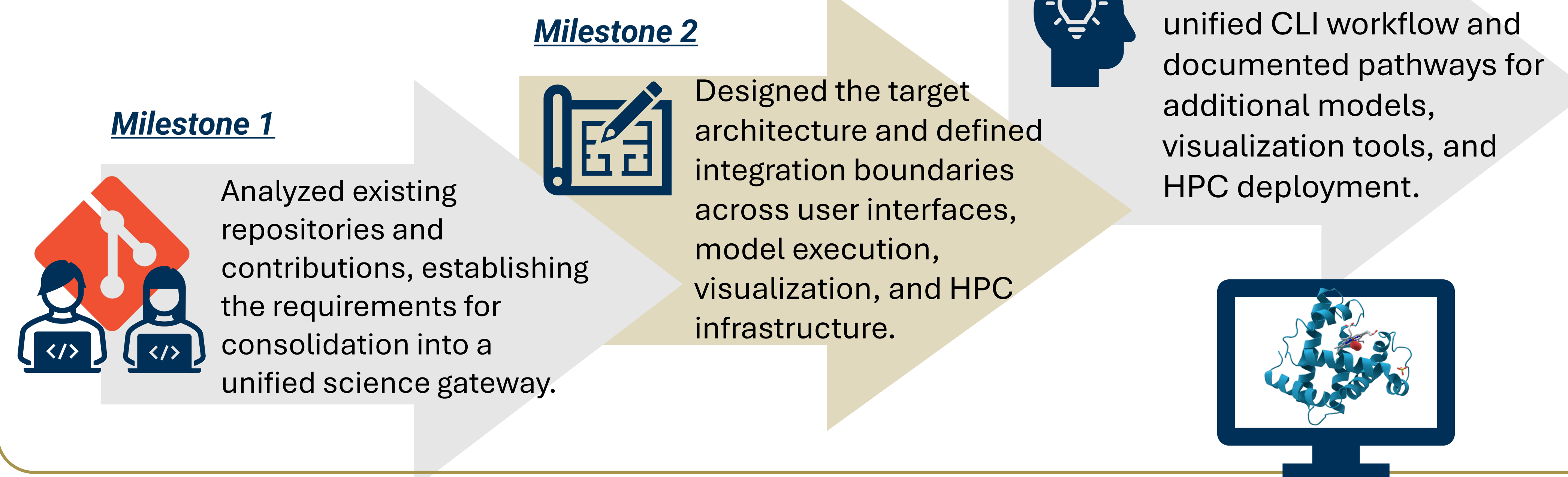


VizFold is an open-source framework for collecting, analyzing, and visualizing intermediate states from protein structure prediction models such as OpenFold, Boltz, and ESMFold.



The project is expanding into a science gateway that provides a consistent foundation for **multi-model execution**, **artifact management**, and **interpretability-driven analysis**.

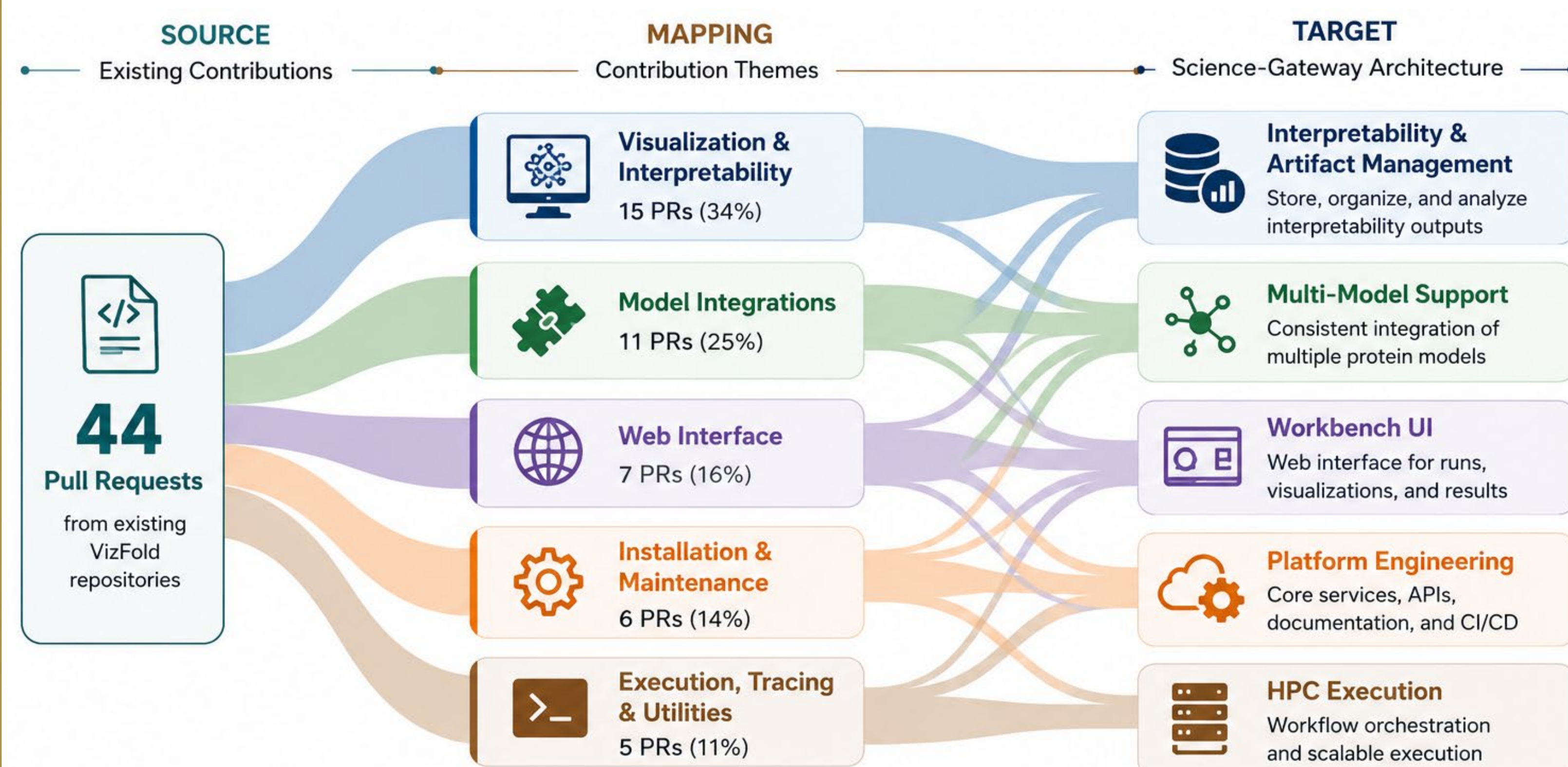
Goals and Milestones



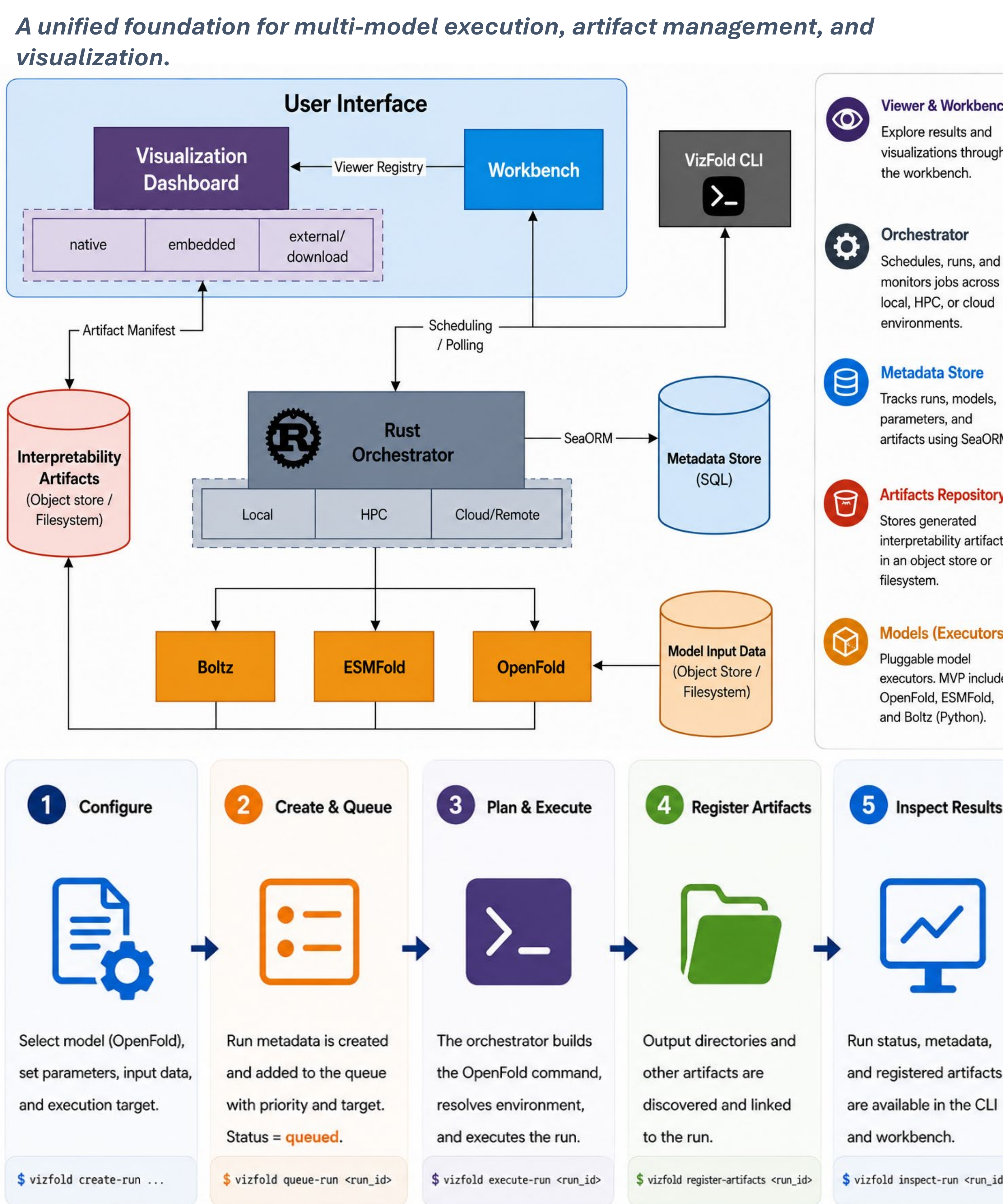
Open-Source Outcomes

- Extended the VizFold platform with a Rust-based OpenFold execution workflow, including preflight validation, command planning, and artifact generation.
- Developed a Next.js visualization dashboard for exploring OpenFold prediction artifacts and attention map outputs.
- Contributed reusable, extensible, open-source software that improves the interpretability and accessibility of protein-folding AI models.

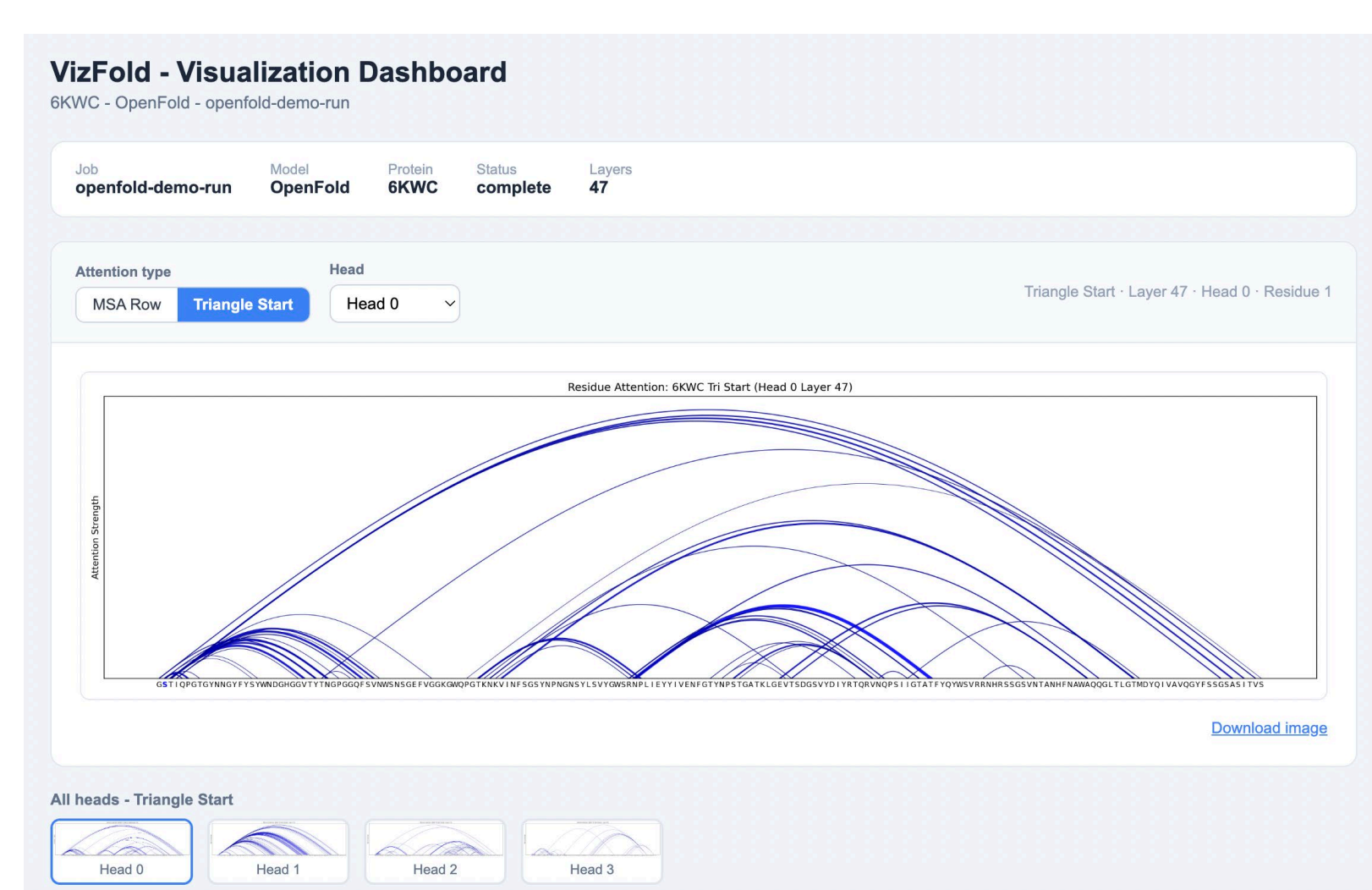
Highlights and accomplishments Repository Inventory



Highlights and accomplishments MVP Architecture and Execution Flow



Highlights and accomplishments Visualization Dashboard



- Toggle between MSA Row Attention and Triangle Start Attention
- Created with Next.js
- User-friendly interface

The dashboard dynamically loads artifacts produced by the Rust executor and presents them through an interactive interface with controls for selecting attention type, layer, and attention head. It allows researchers to explore the internal attention patterns of protein structure prediction models directly in a web browser. The attention scores generated by OpenFold are rendered as arc diagrams, where curved lines connect residue pairs and line thickness encodes attention strength.

Future Work

- Interactive 3D structure viewer: the PDB structure files are already being produced by the executor but currently only offered as a download. Embedding Mol* would let researchers rotate and inspect the predicted 3D structure alongside the attention arc diagrams.
- Additional model backends: the architecture is designed to be model-agnostic, but currently only OpenFold is supported. ESMFold and Boltz-2 backends are the next integration targets.
- CyberShuttle/HPC integration: local execution works for the demo, but connecting the executor to PACE ICE via CyberShuttle would enable full-scale runs on real proteins without GPU hardware on the researcher's machine.
- Science Gateway deployment: moving from a locally-run prototype to a publicly accessible hosted gateway, enabling researchers at other institutions to use VizFold without any local setup.

